Lesson plan · for the teacher

What is data science?

Data science · Level 1 · Lesson 1

Time
40 min

Materials
Board and chalk or markers; Last month's register; A phone or tablet with Darsi, for the lab

Learning goals

- Say what data science is.
- Follow the loop: ask, collect, explore, explain.
- Turn a vague question into a measurable one.

Activity: From a guess to an answer

1. Warm-up (5 min): Ask for guesses about why pupils are late and write them on the board. Point out that these are guesses, not answers.
2. Teach (8 min): Draw the loop: ask, collect, explore, explain. Use the register as the data, and show how a pattern by day leads to a new question.
3. Count (15 min): Groups take one week each of last month's register (no names, just counts) and tally late arrivals by day. Combine the groups' counts on the board in one table.
4. Explain (7 min): Draw a bar chart from the table. Ask: which day stands out? What could explain it? What should we count next?
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence, with a number in it. Then give the exit ticket.

Use counts only; never show which pupil was late.

Discussion

- Which guess on the board did the data support?
- What would we need to count to check the bus idea?
- Which school question would you ask next?

Exit ticket

- Write one measurable question about our school, and the record that would answer it.

Try it in class tomorrow
Ask your pupils for questions about the school that data could answer. Choose one you can measure, and plan what to count this week.

Worksheet · for pupils

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Name _____ Class _____ Date _____

1 **We explained our finding. What comes next in the loop?**

◯ Stop: we are done ◯ A new question ◯ Delete the data

2 **Data scientists mainly build the databases that store a school's data.**

◯ True ◯ False

3 **Put the data science loop in order.**

☐ Explain the finding, then ask again

☐ Explore with numbers and a chart

☐ Collect the data it needs

☐ Ask a clear question

4 **The list holds the days on which pupils arrived late last week. Finish the program so it prints how many late arrivals there were on Monday.**

```
late = ["Mon", "Tue", "Mon", "Wed", "Mon", "Fri", "Mon"]
monday = 0
for day in late:
    pass
print(monday)
```

Write your program:

5 **Match each school question to the data that answers it.**

Which days do pupils arrive late?	• • • •	Marks for each question
Which books are borrowed most?	• • • •	The rain gauge record
Which month is wettest?	• • • •	Library cards
Which topic was hardest in the test?	• • • •	The register

What is one question about your school that you would like data to answer? Where is the data?